

T-Sat 2A Mission Concept Review (MCR)

Knights Satellite Club

1. Mission Statement:

Introducing project, Tethered Satellite 2 Mission A. In the past models of the tethered satellite project involving the mechanical side there was a focus on the parachute deployment system. As such, for this mission of the tethered satellite project we decided to stay near our roots and continue using a deployment system except with a catch. With our focus being on a separation system between the top panel of the CubeSat and the rest of the housing with safe landing through a parachute.

2. Mission Success Criteria

With creating such an unprecedented mission for this version of the tethered satellite, analyzing the mission success involves multiple components to work hand-in-hand without fail. With the greatest achievement being able to remotely control the four servos that will unlock the separation of the systems while having real-time altitude readings and logged pictures through an SD card.

3. Objectives

In creating a 2U CubeSat that would have full control on landing location, it not only gives the Knight's Satellite Club members experience in creating a controlled separation system. Whether the members involved are experienced or inexperienced in designing, prototyping, assembling, or launching a CubeSat. It becomes a perfect introduction to the vast world of CubeSat technology and the engineering behind it. With specifically this mission containing a controlled separation system, it allows for future incorporation of this system into other related projects created by the Knight's Satellite Club.

4. Instrument Payload

T-Sat 2A will incorporate two separate systems, with system one being a 2U CubeSat that contains all the electronics, hardware, and parachute. With the second system being the top panel that will be connected to the balloon. That are connected until desired height is achieved to separate system one into free fall allowing a controlled and safe landing.

System 1

To be more specific, how system one will contain all the hardware involved. First there will be three tiers to this CubeSat. With the bottom tier being the size of a 1U satellite that is going to contain two cameras with one facing straight down through the bottom panel and the second camera being mounted on a side panel to get a side view. In doing so, there will also be a 3-layer PCB stack that will be held on by using the side panels as mounting points for the railing system, which will be used to insert and take out the three layer PCB stack. The PCB stack will be in the assortment of the first layer containing the PCB that will have an ESP32 that will communicate to the camera and SD modules to take and save the pictures. With having the ready

to launch switch, remote control module, attitude module, possibly an acceleration module and any other module added in this PCB layer. With the second layer of the PCB stack will just be used to move around the correct voltage to each respective module and inserts for power, to use both cameras and the four servomotors that will be used. (more in depth explanation of this system in system 2) Finally, the third layer of the stake will just be a 3D-printed layer that will be used to hold down all the batteries. With just using a 3D-printed bar as an anchor to secure the batteries in place. That would summarize what will be in the first tier of the 2U satellite. Leaving there to be 1U area being used for a parachute cavity and the separation system. Where the parachute cavity would be a cubed box that contains a baseplate so the parachute can be neatly folded ready for deployment. Where there will be a few holes so there can be wiring passing through that will be needed to operate the four servos that will be mounted on the top of the walls of the 1.5U satellite so the separation of the two systems can happen. Which will be explained further on how exactly it works, and the attachments need to allow separation in system two.

System 2

As mentioned beforehand there will be a separation created between the two systems, in doing so system two is created where it is mostly just the cover panel the of the satellite and the mounting point from where the helium ballon will be connected to the CubeSat. In creating the separation, which would be where the four servos will take place and be the stabilizing and mounting point for both systems. Each servomotor being mounted on each respective side panel with having an attachment. With the attachment that will be connected to the servomotors being a half cylindrical ring. With having a flat part that can be used to screw into the servomotors while the side that would involve having contact with the part connected to top panel of the CubeSat. Except instead of a complete ring, there will be an opening on the ring. With the part that would be connected to the roof of system two being a hook-like mechanism. The design would involve making it a hook, except it is hanging upside down from the roof. In doing so the servomotor ring link would just be connected to the hook by inserting it through the opening of the ring then, rotating the servo would release the connection to the ring itself. Once commanded or desired altitude has been achieved and through confirmation of the team at the launch sight the servomotor will rotate to the opening in the ring allowing system two to become its own payload which goes into freefall and deploys the parachute through air resistance.

5. EDL Sequence

In the entry, descent and landing sequence would be separated into five different states that will take place as long as the right conditions are made.

Calibration:

Calibration begins once the ready to launch pin is pulled which all power and command to flow through all the modules. Since if the pin is never removed there will be no power or any action being performed on the CubeSat.

Preflight:

Through preflight is a waiting step that involves sensing any sort of gain in altitude to proceed to the next step.

Ascend:

In this stage of the sequence is where the altimeter will be being to take place and give the team back in the launch sight live altitude readings. By this time, the two cameras should have taken pictures, and all other operations are ready and on standby by.

Freefall:

Once this sequence has been promoted the servos will receive a command to create the separation between the two systems allowing the 2U CubeSat to begin freefall and deploy the parachute.

Landing:

This sequence will be working hand-and-hand with the freefall sequence since almost instantly the parachute will catch and the team will be able to see the descending CubeSat with all the pictures and data reading being logged into an SD card.

6. Scientific Values

When involving a parachute deployment, the amount of mass that can be supported for slow descending through the parachute is an important value, since it can vary especially high the load. With also the power of the servos needed to create the swaying and releasing system needed to be calculated to verify it created enough torque to even create rotation.

7. Risk Management

With this mission of T-Sat 2A being the first of its kind that the Knight's Satellite Club has done especially with the separation system, it is bound to have some unrepresented troubles during the creation of project. Nevertheless, there have been a few scenarios that have been deuces that could lead to a mishappen in the project or even the whole deployment. The main concern that is involving this project is the load the servomotors will have to take on. Since all four servos in the separation system will have to take amount the weight of the 2U CubeSat but also the force that would be pushing up to it from the helium balloon itself and any other force that could be inflicted from the ascend whether its from pulling or twirling the rope that links the helium balloon and the CubeSat. Ways we have found that could combat such situations is finding powerful enough servos that can still be operated on at such pressure. The four servos that

oversee the separation system (as explained earlier on how it works) a way to put less strain on the servomotors themselves is having varied materials for the linkages. Instead of creating a 3D-printed part for the hook it can be a metal hook making it not only stronger but also easier to slide against the ring since, the friction against it would drastically less. Involving the parachute on being able to catch the air resistance at the height since there will be no push out system a way the team has thought of combatting this possible problem is having a miniature amount of some type of adhesion between the top panel and the parachute to make sure the parachute is able to get out of the cavity.